



Who Can Assert Ownership Over Automation? Workplace Technological Change, Populist and Ethno-nationalist Rhetoric, and Candidate Support

Sophie Borwein¹ · Bart Bonikowski² · Peter Loewen³ · Beatrice Magistro⁴ · Blake Lee-Whiting³

Accepted: 3 January 2024 / Published online: 3 February 2024

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract

Technological change has dramatically reshaped labor markets, but technology's distributional implications have been less politicized than those of other economic shocks. As innovation in automation and artificial intelligence accelerates, however, political parties face growing incentives to “claim” the issue. Candidates proposing to protect workers against technological change may appeal directly to workers' economic concerns, but if technology is perceived similarly to other economic shocks, workers may also be mobilized through populist and ethno-nationalist appeals. This paper asks: relative to other shocks, what kind of messaging around workplace technological change resonates most with workers? We examine this question using a ten-country 3×3 survey experiment that randomly exposes each respondent to one of three vignettes featuring excerpts from political candidates' speeches: the passages promise to protect workers against offshoring, automation and AI, or changing consumer demand and do so using populist rhetoric, populist and ethno-nationalist rhetoric, or no additional messaging. We find that overall support for protecting workers against technology is lower than for other shocks, but that all types of candidate messaging appeal to workers who feel vulnerable to technology.

Keywords Workplace technological change · Automation · Populism · Ethno-nationalism · Candidate support

Introduction

Technological change is transforming labor markets, with recent advances in robotics, digitization, and artificial intelligence (AI) promising to replace humans with machines for a growing range of workplace tasks (Acemoglu & Restrepo, 2022; Brynjolfsson & McAfee, 2011; Frey & Osborne, 2013). Experience from past automation tells us that even though the benefits of technological adoption

for labor markets may be significant, they are unlikely to be equally distributed. Over the last three decades, a substantial share of skilled workers experienced steady economic gains from technological change, but workers in routine jobs concentrated in the mid-to-low end of the income distribution also lost out in the process (Gallego et al., 2022). Consistent with this trend, technological change has been identified as the largest contributor to growth in income inequality across advanced industrial countries in recent years (Acemoglu & Restrepo, 2022).

Considering the magnitude of labor market disruptions associated with new technologies and findings of significant effects of other economic shocks on public attitudes and behavior (Margalit, 2019), understanding the political implications of workplace technological change has become an important task for social scientists. Research has now established that workers in occupations at objectively higher risk of automation exhibit greater support for populist radical-right parties (Anelli et al., 2021; Gingrich, 2019; Im et al., 2019; Kurer, 2020; Milner, 2021). Why this is the case, however, is less clear. Indeed, recent research suggests that this relationship may have little to do with workers' direct experiences with technological change; while voters who subjectively see automation as a societal threat disproportionately support the radical right, voters who see automation as personally threatening do not (Borwein et al., forthcoming). This raises the possibility that populist-right parties may be exploiting more diffuse fears of economic decline among exposed workers, or redirecting these workers' anxiety toward other more salient targets of blame, such as trade or migration (Gallego & Kurer, 2022; Mutz, 2021; Wu, 2021).

To address these puzzles, our paper asks: who can assert political ownership over the issue of workplace technological change? Political parties have thus far rarely primed technological change (König & Wenzelburger, 2019), or have preferred to misattribute its labor market consequences to other factors such as trade and immigration (Wu, 2021). However, if the pace and scale of technological adoption in labor markets intensifies as some predict, parties and candidates will have growing incentives to politicize the issue (Gallego & Kurer, 2022). The success of populist radical-right parties in capturing the support of workers who feel vulnerable to other sources of economic disruption, such as globalization, raises the question whether the political effects of technological change are different than those of other structural economic transformations. On the one hand, trade and immigration may be more likely than automation to trigger people's inclination to partition the world into in-groups and out-groups, because they involve more easily identifiable "others" who can be blamed for economic turmoil—namely, foreign countries and migrants. Consequently, the populist and ethno-nationalist rhetoric employed by radical-right parties may appeal more to workers exposed to globalization than to technological change (Mutz, 2021). On the other hand, recent experimental research suggests that anxiety about automation can induce anti-immigrant sentiment as well, by intensifying perceptions of economic competition with immigrants. Consistent with this argument, exposing workers to information about technology's potentially negative effects on jobs can cause them to express stronger trade restrictionist and anti-immigrant attitudes (Di Tella & Rodrik, 2020; Gamez-Djokic & Waytz, 2020). This research raises the possibility that populist and ethno-nationalist rhetoric, which has

effectively exploited popular dissatisfaction with trade and immigration, may also appeal to those affected by technological change.

To test if candidates can effectively use populist and ethno-nationalist rhetoric to capture the issue of workplace technological change, this study uses a 3×3 survey experiment fielded in ten countries spanning North America, Europe, and Australia. Respondents are randomly exposed to vignettes describing the potential consequences of one of three different economic shocks: offshoring, automation and AI, or changes in consumer demand. We then evaluate respondents' support for a candidate who proposes to protect workers against these economic changes using populist rhetoric, populist and ethno-nationalist rhetoric, or no additional messaging. We first determine whether workers' candidate preferences differ across the three types of economic shocks. We then examine sub-group variation, focusing on four groups: those subjectively or objectively threatened by automation and AI, and those who report high levels of pre-existing populist or nativist sentiment. By making use of an experimental design, we are able to estimate people's reactions to automation and AI as compared to other economic shocks, which is important because it is challenging to disentangle the labour market effects of automation and shocks such as trade using observational data (Rodrik, 2018).

Our findings are as follows. First, respondents are generally less supportive of candidates who offer protections against automation and AI as compared to offshoring or changing consumer demand. Second, a subsequent sub-group analysis demonstrates that higher perceived threat from automation is associated with stronger support for candidates who target automation and AI, whereas higher objective risk based on respondents' occupation is not. However, subjectively threatened workers also exhibit stronger support for most other candidates and platforms—i.e., they are more supportive of candidates who propose to address not just automation and AI, but also demand changes and offshoring, irrespective of whether candidates feature populist, populist and ethno-nationalist, or no additional messaging in their platforms.¹ Put differently, while workers who report low subjective threat of automation and AI penalize candidates who attach ethno-nationalist appeals to economic platforms as compared to candidates who do not, respondents who report high subjective threat express equally strong support for all candidates. Third, we find that it is not workers exposed to automation and AI, but rather those with high levels of pre-existing ethno-nationalist sentiment, who are most attracted to campaigns that frame automation and AI in ethno-nationalist terms.

Taken together, our findings suggest that protections against automation and AI are less appealing to workers than policy proposals targeting other economic threats. Respondents are also less receptive to the ethno-nationalist messaging than to alternative framings of economic shocks. Nonetheless, candidates can gain advantage among some groups by employing ethno-nationalist messaging: workers who feel subjectively threatened by automation and AI may not be uniquely attracted to the

¹ The exception is for candidates who propose to address offshoring with neutral economic or populist messaging; these candidates are strongly supported by workers with both low and high subjective automation threat perceptions.

ethno-nationalist framing of the issue, but such messaging mobilizes them as much as the other automation and AI frames examined in this study. Thus, there is little cost to candidates' use of ethno-nationalist rhetoric among subjectively threatened workers (as compared to other messaging), while at the same time, such rhetoric has the potential to increase support for automation and AI protections among ethno-nationalist and populist workers, to levels on par with candidates seeking to protect workers against offshoring. This suggests that candidates who employ ethno-nationalist framing of automation and AI could expand their appeal, should the threat of workplace technological change increase among a larger segment of workers.

The Political Implications of Exposure to Workplace Technological Change

Robotics, digitization, and artificial intelligence are unsettling labor markets by facilitating the automation of an increasingly wide range of routine and non-routine workplace tasks (Acemoglu & Restrepo, 2022; Brynjolfsson & McAfee, 2011; Frey & Osborne, 2013). A growing literature seeks to understand the political implications of this technological change. Drawing on the political economy literature on other economic shocks (for a review, see Margalit, 2019), one possibility is that workers exposed to technology will exhibit greater support for the political offerings of left-wing parties. This expectation fits with previous research that finds that workers at higher risk of unemployment increase support for redistributive policies for self-interested material reasons—because these policies protect them in case of income or job loss (e.g., Iversen & Soskice, 2001; Rehm, 2011). Indeed, several studies have found that employment in occupations exposed to automation is associated with voting for mainstream left parties (Gingrich, 2019) and greater demand for redistribution (Thewissen & Rueda, 2019).

Despite the intuitive link between vulnerability to workplace technology and left-wing politics, however, existing research offers more consistent evidence that workers exposed to automation gravitate toward populist radical-right parties. Studies focusing on “survivors” of automation—routine workers who remain employed in their jobs but face the prospect of technology-related unemployment—demonstrate that these workers tend to support European right-wing populist parties (Kurer, 2020), particularly when they report financial hardship (Im et al., 2019). This is corroborated by a number of other studies that document a relationship between individual exposure to automation and radical-right support (Anelli et al., 2021; Gingrich, 2019; Im et al., 2019; Milner, 2021; Kurer, 2020).

While it now seems well established that workers in occupations exposed to automation gravitate toward populist radical-right parties, the mechanisms linking these variables are still unclear. One prominent theory suggests that, among affected workers, technological change and associated economic transformations tend to be interpreted as threatening to their privileged position in the social hierarchy. When combined with other cultural changes, including shifts toward greater gender and racial equality, the perception of economic threat has fostered perceptions of status decline, particularly among low-educated white men (Gidron & Hall, 2017). This

research suggests that even as workplace technological change threatens workers' material well-being, the threat it poses to their psychological well-being may be even more consequential (Kurer, 2020).

Radical-right parties are thought to appeal to workers threatened by workplace technology because they offer platforms that promise to rectify both material and psychological threats (whereas the left prioritizes material threats) (Kurer, 2020). They do this by combining populist and nationalist rhetoric, making vague promises to protect workers against “exogenous changes to the labor market,” which include globalization, migration, and other transformations (Im et al., 2019, p. 2). This form of political discourse often vilifies ostensibly corrupt political elites and ethnic, religious, and racial minority groups, while making nostalgic appeals to a traditional way of life that has been allegedly lost amidst rapid social change (Bonikowski, 2017; Gest et al., 2018).

Thus, research convincingly demonstrates that objective vulnerability to automation is connected to radical-right voting, in part through perceptions of status threat. However, it is unclear whether workers in occupations threatened by technology actually recognize they face this threat. Research examining objective exposure to automation, often using measures of the routineness of respondents' self-reported occupations or other similar measures, finds strong evidence of a relationship between automation exposure and radical-right politics (Anelli et al., 2021; Gingrich, 2019; Im et al., 2019; Milner, 2021; Kurer, 2020), whereas studies measuring workers' subjective perceptions of personal automation threat do not (Borwein et al., forthcoming). These conflicting findings suggest that automation threat may cultivate radical-right voting by instilling in exposed workers a broad sense of economic threat, even though that threat is not specifically attributed to technology. Put differently, workers exposed to automation likely perceive that their occupations are in decline, and therefore that their skills are decreasing in value, but they do not correctly identify technology as the source of this problem (Gallego et al., 2021). It is this broader sense of economic decline—not necessarily connected in voters' minds to technology—that heightens perceptions of status threat that foster radical-right voting.

More recently, research has investigated the possibility that not only do voters fail to attribute automation-related economic decline to technology, but they may instead misdirect blame toward other economic forces, such as trade and immigration (Wu, 2021). As Gallego and Kurer (2022) argue, although economists think that automation has caused more substantial job losses than trade in manufacturing (where job loss has been most visible), separating the labor market effects of automation from globalization is challenging. It therefore seems unreasonable to expect voters to be able to distinguish between the two sources of economic disruption (see also Mutz, 2021; Rodrik, 2018; Wu, 2021). Moreover, there are psychological reasons why voters may prefer to blame economic decline on trade or migration over automation. On the one hand, voters are often quite favorably disposed toward technology, which they associate with positive lifestyle and economic benefits (Gallego et al., 2021; Mutz, 2021; Wu, 2021). On the other hand, immigration and trade are associated with identifiable out-groups—immigrants and foreign countries—that can be easily blamed for technology-induced unemployment. This causal narrative also entails

straightforward policy solutions, such as tariffs and border restrictions (Gallego & Kurer, 2022; Kurer, 2020; Mutz, 2021). The greater ease with which global integration and immigration, as compared to technology, can be villainized may also explain why populist radical-right parties appear to prefer to connect economic grievances to immigration and trade than to automation (Gallego & Kurer, 2022).

Contributions: What if Workplace Technological Change is Made more Salient to Workers?

Taken together, past research makes clear that workers in occupations most exposed to technology are also most likely to support populist and radical-right parties, but this relationship may be unrelated to workers' perceptions of workplace technology per se. Instead, it appears that radical-right parties have capitalized on a broader sense of status threat felt by workers exposed to technology, in part by blaming economic turmoil on trade and immigration, phenomena that trigger stronger emotional responses and perceptions of out-group threat than automation itself. Thus, despite the observed link between automation exposure and right-wing populist voting, it is unclear how the threat of technology specifically—particularly if made salient by politicians—relates to party support. To date, protection against technological change has not been “claimed” by any party family (Gallego & Kurer, 2022). Few political platforms have engaged with the issue of technology at all, and when they have, it has been mostly in a positive light (König & Wenzelburger, 2019). We do not know if the relationship between automation threat and radical-right voting would persist in a world where politicians explicitly discussed the threat of technological change. If automation-exposed workers support radical-right parties primarily because these parties connect their plight to globalization or immigration, would we observe different voting behavior if technological change were made more salient to workers? Or do automation and AI represent a fundamentally different structural change than other economic transformations?

Research focusing on subjective perceptions of—rather than objective exposure to—technology suggests several possible scenarios for how parties could benefit from politicizing technological change. There is mixed evidence that workers who perceive automation and AI as a personal threat exhibit greater support for welfare state policies associated with left-wing parties (Busemeyer & Tober, 2022; Busemeyer et al., 2022; Gallego et al., 2021; Jeffrey, 2021; Kurer & Hausermann, 2022). Subjective automation exposure is related to support for income protection policies, such as enhanced unemployment benefits, but it does not appear to consistently cause workers to support other social spending, such as on pension benefits or active labor market policies (Busemeyer et al., 2022; Kurer & Hausermann, 2022). In addition, although there is little research examining how subjective exposure to automation relates to actual vote choice, correlational evidence from 15 European countries shows that workers who express anxiety about automation-related unemployment tend to be more supportive of left-wing parties (Borwein et al., forthcoming).

Other research suggests, in contrast, that populist radical-right parties may be able to effectively capture automation and AI by appealing to voters' populist and

ethno-nationalist tendencies. Di Tella and Rodrik (2020), for instance, demonstrates that voters exposed to information about technology-induced unemployment exhibit greater support for trade tariffs and other import restrictions, while Gamez-Djokic and Waytz (2020) show that priming voters' concerns about automation increases their support for restrictive immigration policies (see also, Wu, 2022). However, among these studies, only Gamez-Djokic and Waytz (2020) has examined whether support for right-wing politics depends on workers' perceptions of their own vulnerability to automation—and their findings suggest that personal automation threat and anti-immigrant attitudes may not in fact be connected.

In sum, the existing body of research provides suggestive evidence that both left-wing and radical-right parties could benefit from politicizing technology-related labor displacement. We have little direct evidence, however, of what kinds of candidates—or platforms—might capture the support of workers concerned about technology. In a study most closely related to ours, Heinrich and Witko (2022) show experimentally that Americans at low and high risk of automation do not generally express varied preferences for candidates that prioritize different issues, such as the economy or immigration. We ask a related but distinct question: when political candidates choose to actively politicize automation and AI—i.e., when they directly connect the issue of workplace technological change to political rhetoric in their platforms—to whom do they appeal? To answer this question, we examine how supportive respondents are of candidates who propose to protect employees against workplace technology, and how such support changes when candidates frame their proposals in populist and/or ethno-nationalist terms. Furthermore, we examine whether certain kinds of workers find these alternative campaign frames particularly appealing, specifically focusing on those workers who are objectively or subjectively threatened by automation and AI, as well as those who report high levels of pre-existing populist and ethno-nationalist sentiment. Finally, we compare support for candidates who propose to address workplace technological change with support for candidates who use the same frames to address offshoring and demand changes. This analysis allows us to better understand whether workplace technological change is categorically different from other economic shocks (Gallego & Kurer, 2022) or whether workers respond to automation and AI much as they respond to other shocks made salient by politicians.

Data and Methods

To examine the resonance of political messaging around the threat of workplace technological change, we fielded a survey experiment across Europe, North America, and in Australia in the spring of 2022 using online samples provided by the survey firm Cint.² We selected ten high-income countries that vary significantly

² Fielding began on February 14, 2022, with a pre-test survey of 100 respondents in each of the ten countries in our sample. Full fielding of the survey followed from April 26 to June 27, 2022. Due to a translation issue in the French survey, the French pre-test data was not retained in our final sample.

in their welfare state generosity, the extent to which their economies have moved away from employment in routine work, and in the political success of their radical- and populist-right parties. These countries are Australia, Canada, France, Germany, Italy, Poland, Spain, Sweden, the United States, and the United Kingdom. We recruited respondents ages 18 and over, undersampling respondents over 65 to capture a greater proportion of working-age individuals, and administered quotas for age, gender, and region. In analyses, we restrict the sample to respondents who report currently being employed (part-time or full-time), as we are interested in how respondents' own exposure to automation and AI at work shapes their preferences.³ We further exclude respondents with completion times below 10 min, who did not complete the survey, who failed our attention check, and those with duplicate Cint IDs (indicative of repeated survey participation). Approximately 800 to 1000 respondents were retained in each country, yielding a total sample size of 9551.⁴

Survey Experiment

In the experiment, respondents were randomized into three treatment groups, each of which was exposed to a vignette concerning an economic threat facing the country. The threats were framed as having both benefits and costs for the country's economy. They included (1) changing consumer demand, (2) automation and AI, and (3) offshoring. The vignettes read as follows (note that this example is from the U.S. survey, but all prompts were localized):

Demand changes: American companies are responding to changing American consumer demand by adapting their techniques and bringing new products to the market. Companies' adaption to changing consumer preferences is predicted to expand the overall size of the U.S. economy, but some recent economic projections also estimate that a large share of Americans could lose their jobs or face lower wages in the next 5 to 10 years as a result of these changes.

Automation and AI: New advances in automation and artificial intelligence (AI) mean that companies can now use robots and algorithms to perform many jobs that were previously thought to be non-automatable. These advances in automation and AI are predicted to expand the overall size of the U.S. economy, but some recent economic projections also estimate that a large share of Americans could lose their jobs or face lower wages in the next 5 to 10 years as a result of these changes.

Offshoring: Offshoring—where companies move jobs overseas to places like China and India—is an attractive option for American companies looking to lower their production costs and wages. Offshoring is predicted to expand the overall size of the U.S. economy, but some recent economic projections also

³ Online Appendix A.7 shows that core results using the full survey sample are very similar to the worker-only subsample.

⁴ This study is part of a larger study that was pre-registered at OSF: <https://osf.io/5kvn6>.

estimate that a large share of Americans could lose their jobs or face lower wages in the next 5 to 10 years as a result of these changes.

Because we are interested in examining support for political candidates who promise to address the above economic threats, as well as in the efficacy of populist and ethno-nationalist political frames, the economic shock treatments were followed by statements from hypothetical political campaigns. The prompt for these campaign vignettes was as follows:

As you know, from time to time, political parties in the United States replace their party leader. Consider Ethan Adams,⁵ a hypothetical candidate running for his party's leadership on a platform to protect American jobs from exposure to [market changes/automation/offshoring].

One third of respondents was then exposed to a populist message from the candidate, one third to a populist and ethno-nationalist message, and one third saw no further message. The first two treatments were worded as follows:

Populist Message: According to Adams, while Americans increasingly lose their jobs to [market changes/automation, artificial intelligence and robots/offshoring], government has instead been busy serving powerful special interests. As Adams says, “we need to restore politics to the people, so our kids can live the kinds of lives our parents and grandparents lived.”

Populist + Ethno-nationalist Message: According to Adams, while Americans increasingly lose their jobs to [market changes/automation, artificial intelligence and robots/offshoring], government has instead been busy serving powerful special interests, minorities, and immigrants. As Adams says, “we need to restore politics to the people, so our kids can live the kinds of lives our parents and grandparents lived. Back in the day, America was for real Americans, who had deep roots in the country, shared American values, spoke the language, worked hard, and were able to get ahead. It's not like that anymore.”

Finally, respondents were asked about their favorability toward the candidate on a 0–10 scale:

From what you know, would you support Ethan Adams' candidacy?

Other Measures

We anticipate that candidate support may vary among respondents based on their own exposure to workplace technology. To test this possibility, we measured their objective and perceived exposure to technology at work. Our primary measure of objective exposure comes from work by Gallego et al. (2021), and asks respondents about the frequency with which they perform tasks at low risk of automation and AI at work, such as managing or programming (for question wording see Online

⁵ Candidate names were localized.

Appendix A.1). Questions are coded such that higher values denote greater exposure to automation and AI, and the final measure is rescaled to range from 0 to 1. Compared to other measures of objective automation exposure used in past studies, most notably “routine task intensity” (RTI) scores, this approach better captures non-routine tasks that can increasingly be performed by computers and machines (Arntz et al., 2017; Gallego et al., 2021). However, because RTI is commonly used to measure automation risk (Gingrich & Kuo, 2022), we replicate our findings using this method, with the results shown in the Supplementary material. To do so, we assign to each respondent an RTI score based on their self-reported 2-digit ISCO-08 occupation, following the RTI index in Goos et al. (2014).⁶ To measure subjective exposure to automation, we ask respondents if they think that a computer or machine will replace their job within the next five to ten years (response categories: definitely not, probably not, probably yes, definitely yes, don’t know).

We further anticipate that respondents who hold populist and ethno-nationalist beliefs may vary in their support for candidates who frame their economic appeals in populist and ethno-nationalist terms. To measure populist predispositions, we use a battery of six questions developed by Akkerman et al. (2014), which focuses on negative perceptions of political elites as compared to “the people” (see Online Appendix A.2 for question wording). To measure predispositions toward ethno-nationalism, we rely on three questions from the International Social Survey Programme (ISSP) National Identity Supplement, previously used in Bonikowski and DiMaggio (2016). These questions ask respondents about the importance of a person’s ancestry, place of birth, and language fluency for being considered a “true” member of the national community (e.g., truly American) (see Online Appendix A.3 for question wording). These questions tap into respondents’ restrictive notions of who belongs to a nation, often referred to as ethno-nationalism or “exclusionary nationalism” (Bonikowski & DiMaggio, 2016). Both batteries are administered prior to the experimental treatments and we transform the resulting indices to range from 0 to 1. In Online Appendix A.4, we report mean levels of populist and ethno-nationalist sentiment for each country.

Methods

Our analyses use ordinary least squares with observations pooled across countries. We include demographic controls for age group, education, income group,⁷ and gender.⁸ To begin, we examine how information about the three types of economic threats (demand changes, automation and AI, and offshoring) relates to candidate

⁶ Goos et al. (2014) calculated RTI using ISCO-88 codes; we convert ISCO-08 to ISCO-88 codes following Mahutga et al. (2018), and rescale RTI to range between 0 and 1.

⁷ Low-income respondents are those below the 25th percentile of their country’s income distribution in the survey; middle-income are those between the 25th and 75th percentiles; and high-income are those above the 75th percentile.

⁸ Note we did not include country fixed effects in our pre-registration, and we do not include them in analyses. Results are substantively the same when they are included.

support among respondents exposed to populist messaging, populist + ethno-nationalist messaging, and economic messaging alone (i.e., the control condition). We estimate the following equation (Eq. 1), which interacts the economic threat type with the type of candidate messaging:

$$\begin{aligned} \text{candidate.support} = & \alpha + \beta_1 \text{automation}_i \\ & + \beta_2 \text{demand.changes}_i \\ & + \beta_3 \text{populist.message}_i \\ & + \beta_4 \text{natpop.message}_i \\ & + \beta_5 \text{automation}_i * \text{populist.message}_i \\ & + \beta_6 \text{automation}_i * \text{natpop.message}_i \\ & + \beta_7 \text{demand.change}_i * \text{populist.message}_i \\ & + \beta_8 \text{demand.changes}_i * \text{natpop.message}_i \\ & + \beta_9 Z_i + \epsilon_i \end{aligned} \quad (1)$$

where *candidate.support* is the individual respondent *i*'s support for the hypothetical candidate on an 11-point scale, rescaled to range from 0 to 1; *automation* and *demand.changes* are dummy variables indicating the threat to which respondents were exposed (offshoring is the reference group); *populist.message* and *natpop.message* are dummy variables indicating whether the respondent was exposed to the populist or populist + ethno-nationalist message (the lack of additional framing is the reference group); and $\beta_9 Z$ is the vector of demographic covariates.

After estimating our main model, we extend the analysis with four additional equations to examine heterogeneity. The first equation adds an additional interaction for objective automation exposure to the interaction of economic threat and candidate messaging; the second does the same, but for subjective automation threat. Equation three interacts respondents' pre-existing populist attitudes with the economic threat and candidate messaging, whereas equation four does the same for respondents' pre-existing ethno-nationalist attitudes.

Results

Objective and Subjective Threat from Automation and AI

Before analyzing our survey experiment, we provide a descriptive analysis of the relationship between two key variables—objective and subjective (or perceived) threat from automation and AI. Interestingly, subjective and objective threat are essentially uncorrelated (see Online Appendix A.5 for greater detail). Figure 1 shows how respondent demographics relate to the three measures of threat—the task-based measure of objective threat, RTI, and subjective threat. We observe several patterns. Models of objective threat, shown on the left and center of Fig. 1, demonstrate that highly educated and high-income respondents tend to

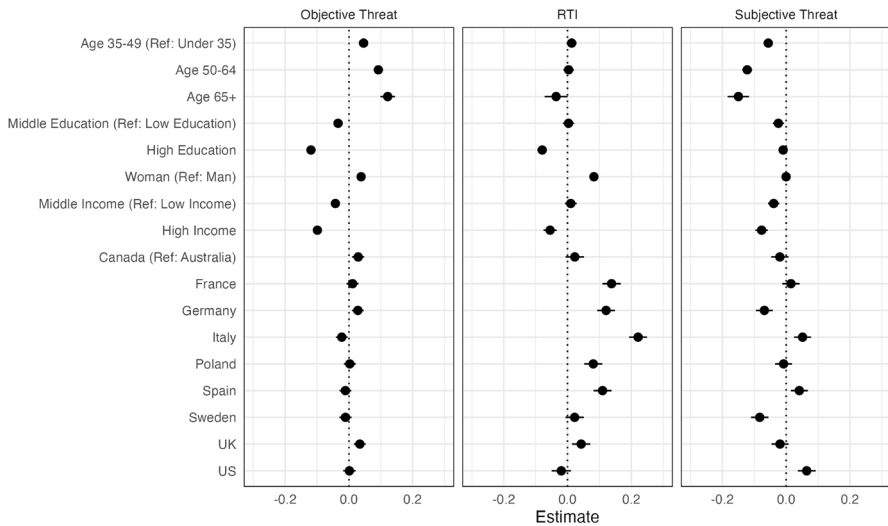


Fig. 1 Sociodemographics and measures of technological threat. Figure shows linear regression coefficients for the relationship between sociodemographic measures and the task-based measure of objective threat (left), RTI (center) and subjective threat (right)

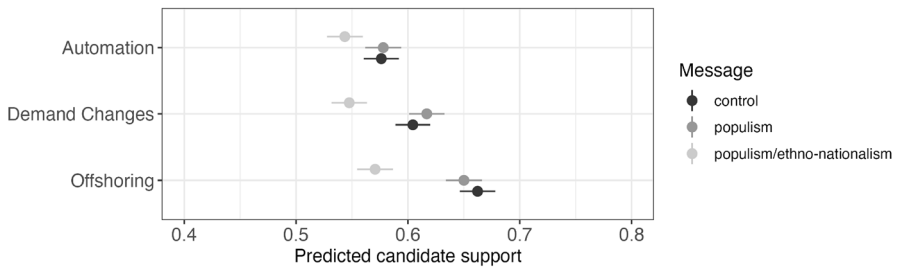


Fig. 2 Predicted candidate support by type of message and economic threat

work in occupations that are at lower risk of displacement by automation and AI. The plot examining subjective risk, on the right, confirms that higher income respondents recognize their lower risk of automation-related job displacement as compared to their low-income counterparts. We further observe that perceptions of exposure decrease with age, even though objective exposure is either unrelated or positively associated with age (depending on whether we use RTI or the task-based measure of risk). One possible explanation for the pattern observed among subjectively threatened respondents is that many older workers in routine-intensive jobs may be holding on to their jobs until retirement (or early retirement) (Kurer & Gallego, 2019) and are therefore less concerned with their job's future. Finally, examining country variation, we note that the U.S. stands out for having

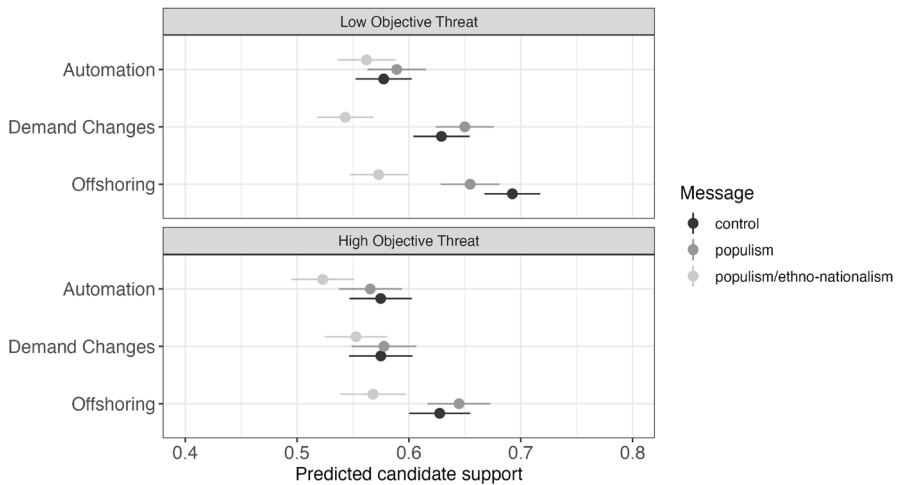


Fig. 3 Predicted candidate support by type of message and economic threat, low and high objective threat

the lowest average RTI score in the survey, even as U.S. respondents report some of the highest levels of subjective threat.

Candidate Support

Figure 2 reports predicted support for a candidate proposing to protect workers against automation and AI, changes in demand, and offshoring disaggregated by the three framing treatments: a populist message, a populist + ethno-nationalist message, or no additional message (see Online Appendix A.6 for regression table). The results demonstrate that support for protecting workers against automation and AI is more limited than support for protecting workers against other economic shocks, particularly offshoring. Focusing on respondents who saw no additional message (the control condition in the second stage of the experiment), we observe that support for the candidate proposing to protect workers against automation is approximately 2.8 percentage points lower than support for a candidate campaigning to protect workers against demand changes, and 8.6 percentage points lower than support for a candidate campaigning to protect workers against offshoring. Online Appendix A.8 further shows that this pattern—of respondents reacting more strongly to offshoring than automation and AI—is consistent across countries, with the exception of the U.S., where both shocks produce similar reactions. Figure 2 also suggests that respondents show some sensitivity to candidate messaging. While candidates proposing to protect workers against automation receive the same level of support regardless of whether or not they use populist framing, candidates who frame their claims in ethno-nationalist terms are significantly less favored by respondents. A similar pattern is observed for demand changes and offshoring.

It is perhaps not surprising that support for protecting workers against automation is moderate, given the significant share of primarily skilled workers who have

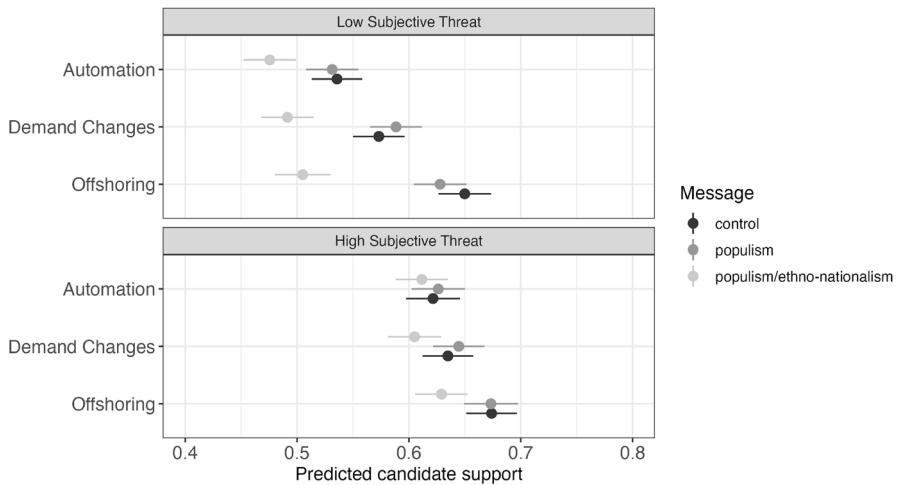


Fig. 4 Predicted candidate support by type of message and economic threat, low and high subjective threat

benefitted from technological change (Gallego et al., 2021). But what about the economic losers of technological change? We might anticipate that these workers will be less sanguine about the benefits of automation and AI and favor candidates who address their concerns. To test this possibility, Fig. 3 examines whether candidate support differs among respondents at low (10th percentile) and high (90th percentile) levels of objective automation risk (see Online Appendix A.9 for regression table). Consistent with the argument that workers at objective risk of automation and AI may not identify technology as a major threat, Fig. 3 shows that respondents at low and high objective automation and AI risk do not significantly differ in their support for a candidate proposing to protect workers against automation and AI. This is true regardless of whether the candidate employs populist framing, populist and ethno-nationalist framing, or no additional framing besides the core economic message.

While workers at objectively high risk of technology-related job loss do not appear to connect their risk exposure to preferences for political interventions targeting automation and AI, it may be the case that these workers are misattributing their potential job precarity to other causes, such as trade, migration, or offshoring (as hypothesized by Mutz (2021) and Wu (2021)), and support candidates whose campaigns confront these shocks. Yet Fig. 3 suggests that respondents at objective automation and AI risk do not differ significantly from low-exposure respondents in their support for candidates proposing to protect workers against offshoring. It should be noted that offshoring is a challenging test of the misattribution hypothesis because most workers report high levels of opposition to offshoring (Wu, 2021).⁹ Regardless,

⁹ In contrast to trade and migration attitudes, Wu (2021) does not find a significant relationship between objective exposure to automation and offshoring attitudes.

respondents exposed to candidate messages targeting offshoring report similar levels of candidate support (or lower, in the case of the control message) irrespective of whether they are objectively exposed to automation and AI risk. As further confirmation, we also find no difference among respondents with low and high levels of objective threat measured using RTI instead of the task-based approach (see Online Appendix A.10).

If respondents objectively exposed to automation and AI do not correctly identify technology-related risks, then it is not surprising that respondents in high and low objective risk occupations hold similar political preferences when it comes to automation and AI. In contrast, we anticipate that workers who perceive technology as threatening to their jobs—regardless of their objective risk—will favor candidates who promise to address the issue. Figure 4 tests this proposition, examining how candidate preferences differ at high (90th percentile) and low (10th percentile) levels of *subjective* threat from automation and AI (see Online Appendix A.11 for regression table). In line with our expectations, Fig. 4 demonstrates that respondents concerned about automation-related personal job loss are significantly more supportive of a candidate who proposes to protect workers against automation than respondents who do not feel at risk. Importantly however, respondents high in subjective risk are responsive not only to candidates proposing to protect workers against automation—they are responsive to candidates proposing to protect workers against all economic threats, including demand changes and offshoring.¹⁰

Interestingly, respondents who feel subjectively threatened by automation and AI are equally responsive to all framing strategies tested in our study, regardless of whether the promise to protect workers against technological shocks is expressed in neutral economic terms, in populist terms, or in populist + ethno-nationalist terms. This finding has several implications. On the one hand, it suggests that respondents high in subjective threat are more amenable to ethno-nationalist messaging than respondents low in subjective threat (the latter are less supportive of ethno-nationalist messaging than of purely economic or populist messaging). On the other hand, it appears that candidates gain little additional leverage with subjectively threatened workers by attaching ethno-nationalist messaging to platforms aimed at automation and AI, because these respondents are just as receptive to alternative framing strategies. Moreover, ethno-nationalist messaging simultaneously repels workers who report low levels of subjective threat, which is likely to dampen overall support for the candidate's campaign (all other things being equal). Finally, Fig. 4 suggests that the most effective strategy for candidates seeking to appeal to workers who perceive high levels of automation threat is to target offshoring; among these respondents, messages focusing on offshoring are slightly more persuasive than those targeting automation and AI.

¹⁰ For the control condition without additional framing, there is no difference in support for the candidate targeting offshoring among respondents with low and high perceived automation threat. Opposition is high in both groups.

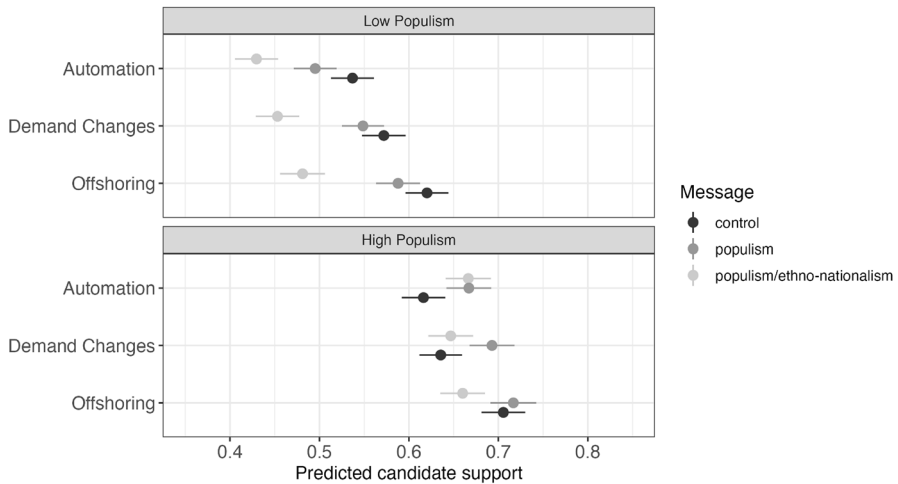


Fig. 5 Predicted candidate support by type of message and economic threat, low and high populism

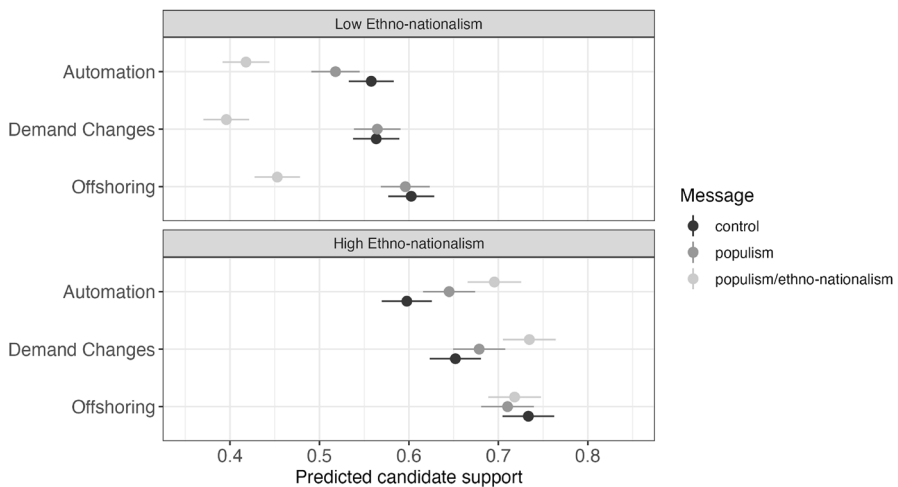


Fig. 6 Predicted candidate support by type of message and economic threat, low and high ethno-nationalism

Populism, Ethno-nationalism, and Candidate Support

Although our findings so far suggest that candidates gain little by framing the threat of automation and AI in ethno-nationalist terms, it may be that ethno-nationalist and populist appeals are more effective among subsets of workers who hold stronger ethno-nationalist and populist attitudes. Previous research suggests that populist and ethno-nationalist attitudes may function as latent dispositions that can become activated through political discourse that heightens perceptions of group threat

(including due to economic shocks), particularly in periods of rapid social change (Bonikowski, 2017).

To test this possibility, we examine how respondents' levels of populism and ethno-nationalism, measured prior to the experimental treatment, predict candidate support. In Fig. 5, we examine candidate support among respondents who report low (10th percentile) and high (90th percentile) levels of populist sentiment, while Fig. 6 examines support among respondents who report low (10th percentile) and high (90th percentile) levels of ethno-nationalist sentiment (see Online Appendices A.12 and A.13 for regression tables).

Figure 5 shows that populism increases support for all three political appeals, regardless of whether they are framed in populist or populist + ethno-nationalist terms. Of particular note, respondents who report high levels of populist sentiment report similar—or higher—levels of support for all candidates than respondents who report high levels of subjective technological threat, as reported in Fig. 4. When exposed to the purely economic platform, both sets of respondents rate the candidate similarly across all three economic threats (automation, demand changes, and offshoring), whereas populist respondents treated with populist and/or populist + ethno-nationalist messages rate the candidate higher than respondents with high levels of subjective threat, most notably when the messaging is attached to automation and AI. Figure 6 further shows that ethno-nationalist respondents are equally or more supportive of all candidates compared to respondents high in subjective threat—and especially so when they are exposed to ethno-nationalist messaging.¹¹ Together these results suggest that in the context of automation, candidates who target workplace technological change and frame their appeals in populist and ethno-nationalist terms can gain greater support from populist and ethno-nationalist respondents than from workers who are threatened by automation and AI (whether objectively or subjectively).¹²

¹¹ Because ethno-nationalism and right-wing ideology are related, in Online Appendix A.14 we also show how candidate support varies among left-, center-, and right-wing respondents. Similar to ethno-nationalist respondents, right-wing respondents find the ethno-nationalist framing of automation equally or more appealing compared to populist and control messaging. In contrast, left-wing respondents are significantly less supportive of candidates who use ethno-nationalist appeals for automation as compared to the populist and purely economic messaging.

¹² A further possibility is that respondents' populist and ethno-nationalist sentiments may interact with their levels of exposure to automation and AI to make certain candidate messages particularly appealing. That is, respondents who are exposed to automation may be particularly susceptible to certain candidate framing strategies when they hold pre-existing populist and/or ethno-nationalist beliefs. Online Appendices A.15 and A.16 examine this possibility, showing how automation threat relates to candidate preferences among respondents who report high levels of populist or ethno-nationalist attitudes (i.e., who are above the median on the respective scales). Among these respondents, feeling subjectively threatened by workplace automation increases support for all candidate messaging targeting workplace automation. As compared to all workers who feel subjectively threatened by automation (shown in Fig. 4), candidate support is greater among subjectively-threatened workers who also report higher levels of pre-existing populist/ethno-nationalist sentiments. Among the subset of populist and ethno-nationalist workers, however, the gap in candidate support between those with low and high levels of subjective threat is similar to the gap in the sample as a whole.

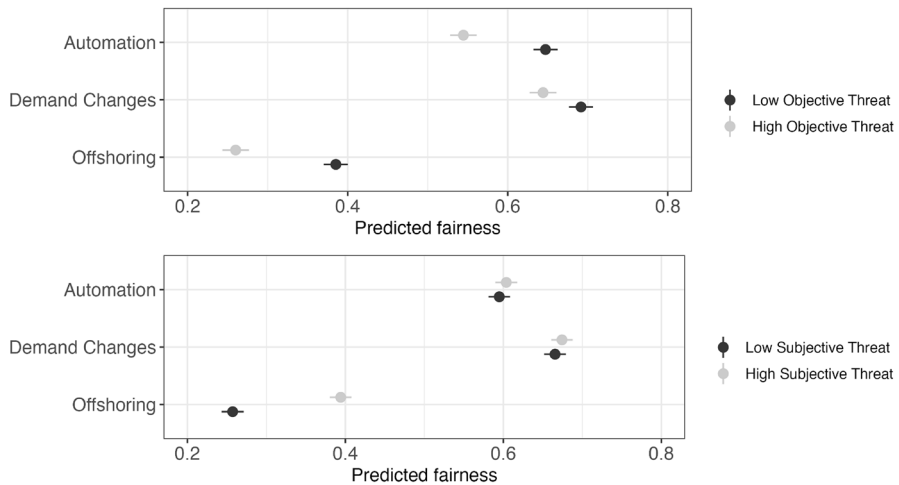


Fig. 7 Perceived fairness of economic threats by low and high objective threat and low and high subjective threat

Extended Analyses: Fairness Perceptions

One particularly notable finding thus far is the lack of an association between objective exposure to workplace technological change and support for candidates proposing to protect workers against automation and AI, whether using populist, ethno-nationalist, or purely economic messaging. These findings are arguably at odds with prior research showing that workers exposed to automation often support populist-right parties and anti-globalization politics, such as restrictions on trade and immigration. Although few studies have explicitly tested the mechanisms that underlie this relationship, a partial explanation has centered around the perceived unfairness of certain economic threats. For example, Rodrik (2018) argues that free trade—often perceived to involve competition with countries that have poorer working conditions and lower wages—is often seen as more unfair than other sources of economic dislocation, such as demand shocks or technological change; such perceptions of unfairness can then stoke political conflict. Others contend that automation, like trade, can also heighten perceptions of unfairness and encourage populist-right voting (Kim & Hall, forthcoming). Because our survey experiment asked respondents about the perceived fairness of automation and AI, offshoring, and demand changes, we are able to assess these alternative scenarios.¹³ In line with Rodrik’s (2018) argument, Fig. 7 demonstrates that offshoring is perceived to be less fair than automation and AI. However, respondents objectively and subjectively threatened by automation differ substantially in their perceptions of the fairness of automation and AI. The top panel of Fig. 7 shows that workers objectively

¹³ Prior to displaying the candidate vignette treatment, we asked respondents: to what extent do you agree that it is fair for [country] companies to [offshore jobs to foreign countries/adopt automation and AI/make changes to adapt to shifts in consumer demand]?

exposed to automation and AI perceive all economic threats—and especially offshoring and automation—to be less fair than respondents who are not objectively threatened. In contrast, the bottom panel of Fig. 7 suggests that respondents who report feeling subjectively threatened by automation do not perceive these other economic forces to be less fair (see Online Appendix A.17 for regression table).

Coupled with our earlier findings, it appears that even though objective exposure to automation contributes to the perception of all economic threats—including those related to automation—as less fair, this perception does not translate into support for candidates promising to protect workers against the same threats. Although we do not have the necessary measures to further explore this finding in this paper, we see several plausible explanations for why objectively exposed workers may perceive automation as unfair, while at the same time withholding support from candidates who seek to protect them against automation and AI's effects. First, studies of income inequality suggest that citizens may express concern about the fairness of rising inequality without seeking concomitant policy responses if they do not trust the government to effectively enact and implement such responses (Kuziemko et al., 2015; Macdonald, 2020). The same may be true of automation and AI-related protections. Alternatively, Kuziemko et al. (2015) and Bartels (2005) also show that publics concerned about inequality often fail to connect their concerns to actual policy responses. A third possibility relates to the design of our experiment, in which the candidate makes only vague promises to protect workers' jobs against exposure to market changes, automation and AI, or offshoring, providing little information about the nature or intensity of the policy response. This lack of specificity may affect respondents' assessments of the value of the proposed political response for the challenge at hand.

In contrast to the results for objective automation exposure, our cumulative findings suggest that respondents who feel subjectively threatened by automation and AI are swayed by candidates proposing to address automation and AI, but not for fairness reasons. While recent research emphasizes the role of fairness considerations in shaping voting behavior, more direct material considerations are also important for understanding economic preferences (Cavaillé, 2023; Kim & Hall, forthcoming; Rodrik, 2018). Thus, uncertainty around job loss associated with automation and AI may heighten workers' perceptions that they are being treated unfairly or it may increase their concern about potential losses of income (whether in relative or absolute terms) (Kim & Hall, forthcoming). Since workers who feel subjectively threatened by automation and AI do not view automation to be less fair than workers who are not subjectively threatened, we suggest that support for candidates seeking to protect workers against automation and AI among the former group may be driven primarily by material rather than fairness considerations.

Conclusion

This study examined how political messaging related to the threat of workplace technological change resonates with workers. To date, workplace technological change has been less politicized than other putative economic threats, such as

trade, offshoring, and immigration. With the effects of technology on labor markets expected to intensify in the coming years, our objective was to understand how politicians may or may not benefit from appealing to workers' anxieties about—and objective exposure to—the adoption of technology in the workplace.

Findings from our survey experiment yield several conclusions. First, we find limited evidence of significant constituencies that are amenable to political messaging that targets workplace technological change. As a whole, workers are less supportive of candidate statements that promise to protect workers against automation and AI as compared to other economic threats, including offshoring and shifts in consumer demand. There is also little evidence that personal exposure to workplace automation and AI increases support for candidates proposing to address the issue. Workers in occupations highly exposed to automation and AI report similar candidate preferences to workers at low objective risk of technology-driven job loss. The subjective perception of being threatened by workplace adoption of technology, meanwhile, is associated with higher support for candidates who propose to protect workers against automation and AI; however, these perceptions are also associated with support for candidates who address all economic threats, at levels that either match or exceed (in the case of offshoring) support for candidates who specifically target automation and AI. This suggests that people who feel threatened by automation and AI may feel a general sense of employment vulnerability, for which they seek political responses.

Our study raises questions about why objective and subjective threat from automation and AI yield distinct political preferences. While objective threat is associated with the view that companies' automation strategies are unfair, it is not associated with preference for candidates promising to protect workers against automation and AI. This pattern is reversed for subjective threat. Following insights from Borwein et al. (forthcoming) and Gallego et al. (2021), we speculate that workers at objective threat of automation and AI may have only a vague sense that their skills and occupations (and by extension, social standing in society) are in decline, and little understanding of who or what is behind their circumstances. Consequently, they may perceive the broader economic system (including automation and AI, offshoring, and demand shocks) as unfair, but they have not formed specific policy demands to address the situation. In contrast, workers who report feeling subjectively threatened by automation and AI are likely to have identified automation as a specific and more imminent material threat, against which they want to be protected through a more immediate policy response.

At the same time, when interpreting our findings it is important to be mindful of the challenges of measuring objective threat of automation and AI. Perhaps most significantly, measures of current exposure cannot account for the possibility that workers may have already been affected by automation, either because they were previously employed in a position that became automated, or because automation has reduced employment opportunities in occupations they might otherwise have entered (Anelli et al., 2021). Moreover, there remains substantial disagreement over how best to capture exposure to automation and AI (Gingrich & Kuo, 2022) and as technology rapidly changes, measures based on attributes of specific occupations

may not keep pace with newly emerging capabilities of computers and machines to automate specific job tasks.

Finally, this study asks if populist and ethno-nationalist messaging might be politically effective in the context of heightened anxiety associated with workplace technological change. To whom do populist and ethno-nationalist frames promising to protect workers against automation appeal? We anticipated that these discursive strategies could appeal to workers who both feel economically vulnerable and report high levels of ethno-nationalism and populism. Among workers who perceive technology as threatening, we indeed find significant support for ethno-nationalist and populist messaging around automation and AI, but these workers are equally supportive of automation-focused political platforms that do not employ populist or ethno-nationalist rhetoric—that is, these workers seek labor market protection and are susceptible to any framing strategy that accompanies promises of economic redress. It is only among workers highest in ethno-nationalism and populism, regardless of whether or not they are personally exposed to automation, that the ethno-nationalist and populist framing of protections against automation encourages greater support.

Coupled with our evidence that respondents concerned about the threat of automation and AI support candidates who make direct economic appeals, but also those who frame their appeals in populist and ethno-nationalist terms, we interpret our findings to suggest that workplace technological change is a political issue that is “up for grabs”—a wide range of candidates and political parties could capture the support of the growing number of workers who are affected by intensifying technological change. At this time, right-wing populist and nativist parties are not necessarily poised to benefit most from the economic dislocation associated with workplace technological adoption, but this may yet change as the political salience of the issue increases in the coming years.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11109-024-09914-0>.

Funding This project was funded with support from Canadian Social Science and Humanities Research Council (IDG 430-2000-498).

Data Availability Replication materials for this study can be accessed at: <https://doi.org/10.5683/SP3/DUCEGX>.

Declarations

Conflict of interest The authors have no competing interests to declare.

References

- Acemoglu, D., & Restrepo, P. (2022). Tasks, automation, and the rise in U.S. wage inequality. *Econometrica*, 90(5), 1973–2016.
- Akkerman, A., Mudde, C., & Zaslove, A. (2014). How populist are the people? Measuring populist attitudes in voters. *Comparative Political Studies*, 47(9), 1324–1353.

- Anelli, M., Colantone, I., & Stanig, P. (2021). Individual vulnerability to industrial robot adoption increases support for the radical right. *Proceedings of the National Academy of Sciences*, 118(47), 1–8.
- Arntz, M., Gregory, T., & Zierahn, U. (2017). Revisiting the risk of automation. *Economic Letters*, 159, 157–160.
- Bartels, L. M. (2005). Homer gets a tax cut: Inequality and public policy in the American mind. *Perspectives on Politics*, 3(1), 15–31.
- Bonikowski, B. (2017). Ethno-nationalist populism and the mobilization of collective resentment. *The British Journal of Sociology*, 68(S1), S181–S213.
- Bonikowski, B., & DiMaggio, P. (2016). Varieties of American popular nationalism. *American Sociological Review*, 81(5), 949–980.
- Borwein, S., Bonikowski, B., Loewen, P., Lee-Whiting, B., & Magistro, B. (Forthcoming). Perceived technological threat, populism, and vote choice: Evidence from 15 European democracies. *West European Politics*.
- Brynjolfsson, E., & McAfee, A. (2011). *Race against the machine: How the digital revolution is accelerating innovation, driving productivity, and irreversibly transforming employment and the economy*. Digital Frontier Press.
- Busemeyer, M. R., Ganderberger, M., Knotz, C., & Tober, T. (2022). Preferred policy responses to technological change: Survey evidence from OECD countries. *Socio-Economic Review*.
- Busemeyer, M. R. & Tober, T. (2022). Dealing with technological change: Social policy preferences and institutional context. *Comparative Political Studies*.
- Cavaillé, C. (2023). *Fair enough? Support for redistribution in the age of inequality*. Cambridge Studies in Comparative Politics Cambridge: Cambridge University Press.
- Di Tella, R., & Rodrik, D. (2020). Labor market shocks and the demand for trade protection: Evidence from online surveys. *The Economic Journal*, 130(628), 1008–1030.
- Frey, C. B., & Osborne, M. (2013). The future of work. *Technological Forecasting and Social Change*, 114(1), 254–280.
- Gallego, A., Kuo, A., Manzano, D., & Fernandez-Albertos, J. (2021). Technological risk and policy preferences. *Comparative Political Studies*.
- Gallego, A., & Kurer, T. (2022). Automation, digitalization, and artificial intelligence in the workplace: Implications for political behavior. *Annual Review of Political Science*, 25(1), 463–484.
- Gallego, A., Kurer, T., & Scholl, N. (2022). Neither left behind nor superstar: Ordinary winners of digitalization at the Ballot box. *The Journal of Politics*, 84(1), 418–436.
- Gamez-Djokic, M., & Waytz, A. (2020). Concerns about automation and negative sentiment toward immigration. *Psychological Science*, 31(8), 987–1000.
- Gest, J., Reny, T., & Mayer, J. (2018). Roots of the radical right: Nostalgic deprivation in the United States and Britain. *Comparative Political Studies*, 51(13), 1694–1719.
- Gidron, N., & Hall, P. A. (2017). The politics of social status: Economic and cultural roots of the populist right. *The British Journal of Sociology*, 68(S1), S57–S84.
- Gingrich, J. (2019). Did states respond to automation matter for voters? *Research & Politics*, 6(1), 1–9.
- Gingrich, J., & Kuo, A. (2022). Gender, technological risk, and political preferences. *Digitalization and the welfare state* (pp. 157–73). Oxford University Press.
- Goos, M., Manning, A., & Salomons, A. (2014). Explaining job polarization: Routine-biased technological change and offshoring. *American Economic Review*, 104(8), 2509–2526.
- Heinrich, T., & Witko, C. (2022). Technology threats to employment, issues, and candidate and party preferences in the United States. *Political Research Quarterly*, 75(3), 797–811.
- Im, Z. J., Mayer, N., Palier, B., & Rovny, J. (2019). The losers of automation: A reservoir of votes for the radical right. *Research and Politics*, 6(1), 2053168018822395.
- Iversen, T., & Soskice, D. (2001). An asset theory of social policy preferences. *American Political Science Review*, 95(4), 875–893.
- Jeffrey, K. (2021). Automation and the future of work: How rhetoric shapes the response in policy preferences. *Journal of Economic Behavior & Organization*, 192, 417–433.
- Kim, S. I., & Hall, P. A. (forthcoming). Fairness and support for populist parties. *Comparative Political Studies*.
- König, P. D., & Wenzelburger, G. (2019). Why parties take up digitization in their manifestos. *Journal of European Public Policy*, 26(11), 1678–1695.
- Kurer, T. (2020). The declining middle: Occupational change, social status, and the populist right. *Comparative Political Studies*, 53(10–11), 1798–1835.

- Kurer, T., & Gallego, A. (2019). Distributional consequences of technological change: Worker-level evidence. *Research & Politics*, 6(1), 2053168018822142.
- Kurer, T., & Hausermann, S. (2022). Automation risk, social policy preferences, and political participation. *Digitalization and the welfare state* (pp. 139–56). Oxford University Press.
- Kuziemko, I., Norton, M. I., Saez, E., & Stantcheva, S. (2015). How elastic are preferences for redistribution? Evidence from randomized survey experiments. *American Economic Review*, 105(4), 1478–1508.
- Macdonald, D. (2020). Trust in Government and the American public's responsiveness to rising inequality. *Political Research Quarterly*, 73(4), 790–804.
- Mahutga, M. C., Curran, M., & Roberts, A. (2018). Job tasks and the comparative structure of income and employment: Routine task intensity and offshorability for the LIS. *International Journal of Comparative Sociology*, 59(2), 81–109.
- Margalit, Y. (2019). Political responses to economic shocks. *Annual Review of Political Science*, 22(1), 277–295.
- Milner, H. V. (2021). Voting for populism in Europe: Globalization, technological change, and the extreme right. *Comparative Political Studies*, 54(13), 2286–2320.
- Mutz, D. C. (2021). Mis-attributing the causes of American job loss. *Public Opinion Quarterly*, 85(1), 101–122.
- Rehm, P. (2011). Social policy by popular demand. *World Politics*, 63(2), 271–299.
- Rodrik, D. (2018). Populism and the economics of globalization. *Journal of International Business Policy*, 1(1–2), 12–33.
- Thewissen, S., & Rueda, D. (2019). Automation and the welfare state: Technological change as a determinant of redistribution preferences. *Comparative Political Studies*, 52(2), 171–208.
- Wu, N. (2021). Misattributed blame? Attitudes toward globalization in the age of automation. *Political Science Research and Methods*, 1–18.
- Wu, N. (2022). Restrict foreigners, not robots: Partisan responses to automation threat. *Economics and Politics*.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Sophie Borwein¹  · Bart Bonikowski² · Peter Loewen³ · Beatrice Magistro⁴ · Blake Lee-Whiting³

✉ Sophie Borwein
sophie.borwein@ubc.ca

Bart Bonikowski
bonikowski@nyu.edu

Peter Loewen
peter.loewen@utoronto.ca

Beatrice Magistro
magistro@caltech.edu

Blake Lee-Whiting
blake.lee.whiting@mail.utoronto.ca

- ¹ University of British Columbia, Vancouver, BC, Canada
- ² New York University, New York, NY, USA
- ³ University of Toronto, Toronto, ON, Canada
- ⁴ California Institute of Technology, Pasadena, CA, USA